Genuine Multipartite Entanglement Analysis

Hubbard ring - Convex-roof framework

System parameters

Model	**Hubbard 1D-PBC**
Mode	**uniform**
Sites (L)	**4**
Electrons (Ne)	**4**
Hopping (t)	**1.0**
Interaction (U)	**4.0**
n-body order	**3**
Eigenstate idx	**0 (ground state)**

Eigenstate properties

Energy E0	**-2.10274848**
Energy E1	**-1.80642385**
Gap (E1-E0)	**0.296325**
<S^2>	**0.000000**
S	**0.0000**
<Sz>	**0.000000**
Multiplicity	**1.0**
Spin label	**Singlet (S=0)**

Double occupancy

<D> total	**0.287325**
<D> / site	**0.071831**

RDM and GME statistics for this run (3-body clusters)

Hilbert space dim : 2^3 = 8
Number of clusters : 0
Pure clusters : 0
Mixed clusters : 0
Clusters with GME (>0) : 0

Purity gamma = Tr(rho^2) : mean = — [min — , max —]
Von Neumann entropy : mean = — [min — , max —]
Rank of rho : mean = — [min — , max —]
C_GME (best estimate) : mean = — [min — , max —]

Summary: All 3-Body RDM Clusters

System: L=4, Ne=4, U=4.0, t=1.0 | E_0=-2.102748

Sites	Purity	C_GME	C_LB	C_cayl	C_expo	C_stiefel	C_UB	Status

Legend:
  Purity = Tr(rho²) [1.0=pure, <1.0=mixed] | C_GME = Genuine multipartite entanglement (pure only)
  C_LB = Lower bound (notes eqs.6-9) | C_cayl = unitary optimization - Cayley Transform |C_expo= unitary optimization - Expoenetial Map | BEST
  C_UB = Upper bound (eigendecomp) | Status: GME =has GME | NO GME=separable | MIXED=mixed state